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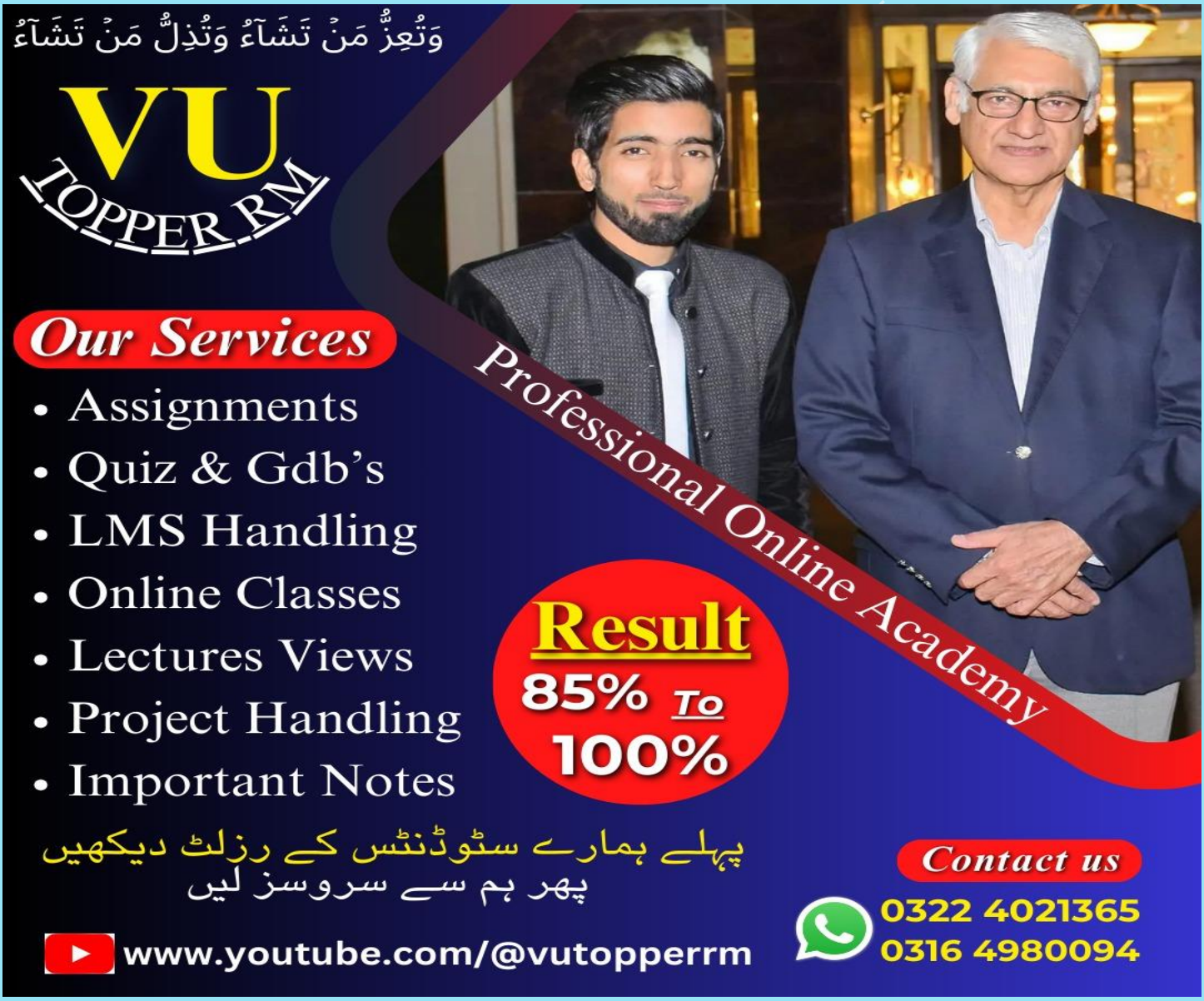
STA301-Statistics And Probability

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If Var(mean1) < Var(mean2) it implies:

A. Mean2 is more efficient
B. Mean2 is more consistent
C. Mean1 is more consistent
D. Mean1 is more efficient

An estimator is said to be efficient if it has

A. Unbiased estimate
B. Smallest Variance
C. Both (a) & (b)
D. None of these

In sampling from a large population with sigma= 12, the standard error of the mean is found to be 3. The size of the sample used is:

A. 16
B. 12
C. 22
D. 11

If the two populations are non-normal and both the sample sizes are large, then the sampling distribution of the difference of means will be:

A. t-distribution
B. Normal distribution
C. Chi-square distribution
D. F-distribution

The standard error of the sampling distribution of differences of means (x 1 - x 2) ,is equal to:

A. √((ϭ1+ϭ2)/n)
B. None of the above
C. √(ϭ/n)
D. √((ϭ1)2/n1)+ (ϭ2)2/n2

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P 0 1 2 3
f(p) 1/20 9/20 9/20 1/20
The mean of the sampling proportion will be:
Select the correct option

A. 10/20
B. 30/20
C. 25/20
D. 15/20

Normal approximation to the binomial distribution is used when:

A. np>5
B. Both of the above
C. None of the above
D. nq>5

A parameter is a ..... quantity.

A. Constant
B. Sample
C. Random
D. Variable

In a sampling experiment the mean value of all the possible values of the statistic was found to be equal to the corresponding population parameter. This condition suggests that the statistic is:

A. A biased estimate for the parameter
B. An unbiased estimator of the parameter
C. Approximately normally distributed
D. Normally distributed

Suppose we have a population of size N = 5 i.e. 0,11,20,23,45 and if we draw all possible samples of size n = 2 with replacement then total possible samples
will be:

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A. 35
B. 50
C. 25
D. 10

In interval estimation we obtained a ……of values as an estimate of parameter.

A. Range
B. Group
C. Both a and b
D. None of these

P 0 1/3 2/3 1
f(p) 1/20 9/20 9/20 1/20
The mean of the sampling proportion will be:

A. 11/20
B. 12/20
C. 9/20
D. 10/20

If an estimator is more efficient then the other estimator, its shape of the sampling distribution will be

A. Normal
B. Flattered
C. Highly peaked
D. Skewed to right

If our sampled population is normal, then sampling distribution of the sample mean will ……:

A. None of the above
B. Depends on sample size
C. Not necessarily normal
D. Normal

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Which of the following statements is correct?

A. A point estimate is an estimate of the range of a population parameter
B. All of the above statements are correct
C. A point estimate is an unbiased estimator if its standard deviation is the same as the actual value of the population standard deviation
D. A point estimate is a single value estimate of the value of a population parameter

If Var(T1) >Var(T2) ,where T1 and T2 are two unbiased estimators ,then:

A. T1 is more efficient
B. T2 is more consistent
C. T1 is more consistent
D. T2 is more efficient

In the method of moments, how many equations are required for finding two unknown population parameters?

A. 1
B. 2
C. 3
D. 4

The sampling distribution of the mean becomes approximately normally distributed only when the following conditions is met?

A. The standard deviation of the population is large.
B. A single random sample is drawn from the population.
C. The sample size is large.
D. The population is normally distributed

We can apply method of Maximum Likelihood on:

A. Qualitative variables only
B. Continuous variables only
C. Discrete variables only

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D. Discrete as well as continuous variables

Sample mean is a/an ..... estimate of population mean.

A. Unbiased
B. None of the above
C. Biased
D. Either unbiased or biased

Method of Moments for estimation of parameters is based on the theory of

A. None of the above
B. Linear Estimation
C. Non-linear estimation
D. Both Linear and Non-linear

Which one of the statement is true?

A. E (S2) ≠ s2
B. E (S2) = s2/n
C. E (s2) = б2
D. E (s2) = б2/n

Which of the following must not be a consistent estimator ?

A. Sample median
B. Sample mean
C. Sample variance
D. Sample Proportion

How can we interpret the 90% confidence interval for the mean of the normal population.

A. There are 10% chances of falling true value of the parameter
B. There are 90% chances of falling true value of the parameter
C. There are 100% chances of falling true value of the parameter
D. All are correct

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In sampling without replacement; fpc is not used when:

A. n=0.05N
B. None of the above
C. n>0.05N
D. n<=0.05N

As the sample size _____ the variation of the sampling distribution of the same means _____:

A. Increases, decreases
B. Increases, remains the same
C. Decreases, decreases
D. Decreases, remains the same

When sampling is from an infinite population, the mean and standard error of the sampling distribution of p^ are _____ and_____, respectively:

A. p, p(1 - p)
B. p^, p(1 - p) / n
C. p, sqrt [p(1 - p)/n]
D. p, n(1 - p)p

Question No:1 **(Marks:1)** **Vu-Topper RM**

Uniform distribution is a:

A. Mixed
B. Discrete distribution
C. Continuous Distribution
D. None of these

Question No:2 **(Marks:1)** **Vu-Topper RM**

A confidence interval has a specified probability_____ of containing the true value of the parameter.

A. α
B. 1- α

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C. 1+ α
D. α/2

Question No:3 **(Marks:1)** **Vu-Topper RM**
When the event occur randomly over a specified interval of time, space or length; the distribution is known as:
A. Poisson Process **Page 217**
B. None of the above
C. Poisson Distribution
D. Hyper geometric distribution

Question No:4 **(Marks:1)** **Vu-Topper RM**
Which of the following is true for the Poisson distribution:
A. mean > variance
B. mean < variance
C. mean = variance **Page 233**
D. mean = standard deviation

Question No:5 **(Marks:1)** **Vu-Topper RM**
Poisson distribution is the limiting form of binomial distribution when n, the number of trials is large and p, the probability of success is
A. very Small **Google**
B. Very large
C. Moderately small
D. Moderately large

Question No:6 **(Marks:1)** **Vu-Topper RM**
The standard normal distribution has it mean and variance equal to:
A. 0 and 0 respectively
B. 1 and 1 respectively
C. 0 and 1 respectively **Google**
D. 1 and 0 respectively

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Question No:7 **(Marks:1)** **Vu-Topper RM**

When an interval is constructed, what is the probability of occurrence of parameter in it?

A. 1
B. 0
C. 50
D. Either 1 or 0 **Page 263**

Question No:8 **(Marks:1)** **Vu-Topper RM**

When we draw the sample with replacement, the probability distribution to be used is:

A. Poisson
B. Binomial
C. Hypergeometric
D. Binomial & hypergeometric

Question No:9 **(Marks:1)** **Vu-Topper RM**

For the Poisson distribution P(X = 1) = e−0.1350.13511! the mean value is:

A. 1
B. 0.135 **Page 222**
C. -0.135
D. e−0.135

Question No:10 **(Marks:1)** **Vu-Topper RM**

A quantity obtained by applying certain rule or formula is known as

A. Estimate **Page 264**
B. Estimator
C. Parameter
D. Proportion

Question No:11 **(Marks:1)** **Vu-Topper RM**

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A random sample of n = 6 has the elements 6, 10,13,14,18 and 20. What is the point estimate of the population mean?

A. 11
B. 12
C. 11.5
D. 13.5

Question No:12 (Marks:1) Vu-Topper RM

An urn contains 4 red balls and 6 green balls. A sample of 4 balls is selected from the urn without replacement. It is the example of:

A. Binomial distribution
B. Hypergeometric distribution Page 219
C. Poisson distribution
D. Exponential distribution

Question No:13 (Marks:1) Vu-Topper RM

A statistic whose standard error decreases with an increase in the sample size will be .....

A. None
B. Efficient
C. Unbiased
D. Consistent

Question No:14 (Marks:1) Vu-Topper RM

The larger the standard error, the ..... the confidence interval.

A. Wider
B. Accurate
C. Inaccurate
D. Narrower

Question No:15 (Marks:1) Vu-Topper RM

The mean deviation of the normal distribution is approximately:

A. 7/8 of the S.D

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B. 4/5 of the S.D
C. 3/4 of the S.D
D. 1/2 of the S.D

Question No:16 **(Marks:1)** **Vu-Topper RM**
Mean and variance of a Poisson Distribution are:
A. Equal **Google**
B. Not equal
C. Mean is greater than variance
D. Variance is greater than mean

Question No:17 **(Marks:1)** **Vu-Topper RM**
If the first moment ratio is greater than 0, then the distribution will be:
Moment Ratios **Page 222**

Question No:18 **(Marks:1)** **Vu-Topper RM**
A statistic whose standard deviation decreases with an increase in sample size will be called:
A. None
B. Efficient
C. Unbiased
D. Consistent

Question No:19 **(Marks:1)** **Vu-Topper RM**
If we have two normally distributed populations then sampling distribution of the difference of sample means will be:
Normal distribution

Question No:20 **(Marks:1)** **Vu-Topper RM**
Sample Proportion is a /an_____ estimator of population mean.
A. Unbiased
B. Consistent
C. None of these

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D. Unbiased and Consistent estimator

Question No:21 **(Marks:1)** **Vu-Topper RM**

The range of the binomial distribution is:

A. 1,2, …….n

B. 0, 1, 2, …… n

C. 0, 1, 2, …… x

D. 0, 1, 2, …… 100

Question No:22 **(Marks:1)** **Vu-Topper RM**

In Binomial distribution the sample size is considered to be sufficiently large, if both np and nq are greater than or equal to:

A. 3

B. 5 **Google**

C. 10

D. 15

Question No:23 **(Marks:1)** **Vu-Topper RM**

The probability distribution of the proportion of successes in all possible samples is called the:

A. Sampling distribution **Page 237**

B. Sampling probability distribution

C. Sampling distribution of sample proportions

D. Sampling distribution of Population proportions

Question No:24 **(Marks:1)** **Vu-Topper RM**

Consider a large population with a mean of 160 and a standard deviation of 20. A random sample of size 64 is taken from this population. What is the standard deviation of the sample mean?

A. 1.654

B. 2.500

C. 3.125

D. 3.568

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Question No:25 **(Marks:1)** **Vu-Topper RM**

providing as an estimate a single value calculated from the sample that is likely to be close in magnitude to the unknown parameter is known as

A. Estimation
B. Point Estimation **Google**
C. Interval Estimation
D. None of the above

Question No:26 **(Marks:1)** **Vu-Topper RM**

The width of the confidence interval is called_____ of the estimate.

A. Biased
B. Accuracy
C. Precision **Google**
D. Unbiasedness

Question No:27 **(Marks:1)** **Vu-Topper RM**

To determine a sample size which test we use …….

A. F
B. Z
C. T
D. None of these

Question No:28 **(Marks:1)** **Vu-Topper RM**

The central limit theorem states that the mean of the sampling distribution of the means is equal to the population:

A. Mean
B. Random Error
C. Standard error
D. Standard deviation

Question No:29 **(Marks:1)** **Vu-Topper RM**

In a Geometric distribution, Maximum Likelihood Estimator (MLE) for

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proportion (P) is equal to:

A. Sample mean
B. Sample variance
C. Reciprocal of the mean
D. Reciprocal of the sample variance

Question No:30 (Marks:1) Vu-Topper RM

Sampling distributions are:

A. None oof the given
B. Refers to the margine of error
C. The probability distribution of sample statistic Google
D. The probability distribution that is assigned to population parameters.

Question No:31 (Marks:1) Vu-Topper RM

In binomial distribution, formula of calculating mean is:

A. U=np
B. U=pq
C. U=qn
D. U=p+q

Question No:32 (Marks:1) Vu-Topper RM

Ideally, the width of confidence interval should be

A. 0
B. 1
C. 98
D. 100

Question No:33 (Marks:1) Vu-Topper RM

The binomial distribution is negatively skewed when:

A. p>q Page 215
B. p=q
C. p<q

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D. p=q=1/2

Question No:34 **(Marks:1)** **Vu-Topper RM**

Interval estimation and confidence interval are:

A. Same
B. Opposite
C. Different
D. Independent

Question No:35 **(Marks:1)** **Vu-Topper RM**

When N becomes indefinitely large (N tends to infinity) hyper geometric distribution follows:

Normal distribution **Page 288**

Question No:36 **(Marks:1)** **Vu-Topper RM**

When the population is normally distributed, population standard deviation is unknown and the sample size is n = 15, the confidence interval for the population mean is based on

A. the Binomial distribution.
B. the z (standard normal)distribution.
C. the Poisson Distribution.
D. the t distribution.

Question No:37 **(Marks:1)** **Vu-Topper RM**

The probability of Type II error is equal to:

A. Alpha
B. 1-Alpha
C. Beta
D. 1-Beta

Question No:38 **(Marks:1)** **Vu-Topper RM**

If we are testing H0 : mean = 250 againstH1 : mean > 250 (exceeds 250).Then rejection region will be

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A. on left side of mean
B. on right side of mean
C. In the center
D. on both sides of mean

Question No:39 **(Marks:1)** **Vu-Topper RM**

The proportion of working females in Pakistan is at most 0.40, the Null Hypothesis is H0 : P≤ 0.40. then the alternative hypothesis H1 is

A. P = 0.40
B. P < 0.40
C. P > 0.40
D. none of these

Question No:40 **(Marks:1)** **Vu-Topper RM**

"The temperatures of patients were checked before and after given a certain medicine." This is an example of:

A. Pairing by design
B. Natural pairing of data
C. Independent data
D. Automatic grouping

Question No:41 **(Marks:1)** **Vu-Topper RM**

S^2 for large samples and s^2 for small samples will be approximately equal when:

A. n is small
B. n is less than 10
C. n is equal to 20
D. n is large

Question No:42 **(Marks:1)** **Vu-Topper RM**

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An experiment was conducted to estimate the mean yield of a new variety of wheat. A sample of 20 plots gave a mean yield of 2.9, and a 95% confidence interval of (2.48, 3.32) . This means:

A. None of these
B. We are sure the true mean yield of this new variety is between 2.48 and 3.32.
C. We are 95% con?dent that the true mean yield of this variety is between 2.48 and 3.32
D. We are 95% confident that the true mean yield of this variety is 2.9

Question No:43 (Marks:1) Vu-Topper RM

The t-distribution can never become narrower than:

A. F-distribution
B. exponential distribution
C. Chi-square distribution
D. standard normal distribution

Question No:44 (Marks:1) Vu-Topper RM

Which one of the following provides the basis for hypothesis testing?

A. Critical value
B. Alternative hypothesis
C. Test-statistic
D. Null hypothesis

Question No:45 (Marks:1) Vu-Topper RM

The larger the n, the smaller the standard error, and so the narrower the

A. None of the above
B. Confidence level
C. Level of significance
D. Confidence Interval

Question No:46 (Marks:1) Vu-Topper RM

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What will be the value of the variance of t-distribution, for v >2 : where v is degree of freedom v=n-1

A. Greater than zero
B. Less than one
C. Equal to one
D. Greater than one

Question No:47 **(Marks:1)** **Vu-Topper RM**

t-distribtion is applicable in case of

A. Independent samples
B. Non-Normal populations
C. Dependent samples
D. Both independent and dependent samples

Question No:48 **(Marks:1)** **Vu-Topper RM**

The precision of an estimator can be increased by decreasing the;

A. Size of sample
B. Size of population
C. Number of parameters
D. Number of estimators

Question No:49 **(Marks:1)** **Vu-Topper RM**

"A failed student was passed by the examiner", is an example of:

A. Type-IV error
B. Type-III error
C. Type-II error
D. Type-I error

Question No:50 **(Marks:1)** **Vu-Topper RM**

When we start hypothesis testing, we always assume that:

A. H1 is true
B. All are correct
C. H0 is false

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D. H0 is true

Question No:51 **(Marks:1)** **Vu-Topper RM**

A null hypothesis is generally denoted by:

A. H0
B. H1
C. H2
D. H3

Question No:52 **(Marks:1)** **Vu-Topper RM**

Suppose H0: u 1 < u 2 H1 :u 1 > u 2 and Critical value is = 1.645 If calculated value of Z = 1 .88 then what will be your conclusion?

A. None of the above
B. Reject Ho
C. Accept Ho
D. Impossible to decide

Question No:53 **(Marks:1)** **Vu-Topper RM**

Which is the most appropriate statistical technique used to test for significant differences between two group means from data gathered by a repeated measures design?

A. None of the above
B. Student's t-test
C. Paired t-test
D. Pearson correlation

Question No:54 **(Marks:1)** **Vu-Topper RM**

If type I and type II errors are fixed, then power of test increase with

A. Increasing the sample size
B. Changing the sample size
C. Decrerasing the sample size
D. None of above

Question No:55 **(Marks:1)** **Vu-Topper RM**

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Inferential statistics involves:

A. All of these
B. Estimation
C. Confidence interval
D. Testing of hypothesis

Question No:56 **(Marks:1)** **Vu-Topper RM**

Conventionally, the probability of making a type-II error is denoted by:

A. ß (beta)
B. Ɵ (theta)
C. a (alpha)
D. б (sigma)

Question No:57 **(Marks:1)** **Vu-Topper RM**

For a sample of size 20 taken from a normally distributed population with standard deviation equal to 5, a 90% confidence interval for the population mean would require the use of:

A. t = 1.328
B. t = 2.12
C. t = 1.729
D. z = 1.645

Question No:58 **(Marks:1)** **Vu-Topper RM**

The level of significance is also called:

A. Acceptence region
B. Power of test
C. Type I error
D. Type II error

Question No:59 **(Marks:1)** **Vu-Topper RM**

The proportion of working females in Pakistan is at most 0.40,the null hypothesis Ho<0.40 the alternative hypothesis H1 is:

A. P = 0.40

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B. P > 0.40
C. P ≠ 0.40
D. P < 0.40

Question No:60 **(Marks:1)** **Vu-Topper RM**
Suppose that the workers of factory B believe that the average income of the workers of factory A exceeds their average income; the null hypothesis will be:

A. MeanA ≤ MeanB
B. MeanA < MeanB
C. MeanA ≥ MeanB
D. MeanA > MeanB

Question No:61 **(Marks:1)** **Vu-Topper RM**
The proportion of cigarette smokers in Pakistan is greater than 20%, the null hypothesis Ho is:

A. P ≤ 0.20
B. P > 0.20
C. P < 0.20
D. P ≠ 0.20

Question No:62 **(Marks:1)** **Vu-Topper RM**
The test statistic to test the U1 = U2 (U represent the mean of population)for normal population when n>30.

A. All of Above
B. T-Test
C. Z-Test
D. F-Test

Question No:63 **(Marks:1)** **Vu-Topper RM**
The degrees of freedom associated with paired t-test ……

A. n1-2 and n2-2
B. n-1

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C. n1+n2-4
D. n-2

Question No:64 **(Marks:1)** **Vu-Topper RM**
As sample size goes up, what tends to happen to 95% confidence intervals?

A. They become more precise and narrow
B. They become more narrow
C. They become wider
D. They become more precise

Question No:65 **(Marks:1)** **Vu-Topper RM**
The confidence intervals become wide and less precise by

A. increasing the sample size
B. decreasing the sample size
C. decreasing the level of significance
D. None of these

Question No:66 **(Marks:1)** **Vu-Topper RM**
For test of hypothesis Ho: $\mu 1 = \mu 2$ and H1: $\mu 1 < \mu 2$,the critical region at 0.05 level of significance and $n > 30$ is:

A. Z less and equal 1.96
B. Z> 1.96
C. Z>1.645
D. Z<-1.645

Question No:67 **(Marks:1)** **Vu-Topper RM**
An industrial designer wants to determine the average amount of time it takes an adult to assemble an "easy to assemble" toy. A sample of 16 times yielded an average time of 19.92 minutes, with a population standard deviation equal to 5.73. Assuming normality of assembly times, what is a 95% confidence interval for the mean assembly time?

A. (14.1123, 22.7277)
B. (17.1123, 24.7277)

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C. (17.1123, 22.7277)
D. (17.1123, 30.7277)

Question No:68 **(Marks:1)** **Vu-Topper RM**
If we are testing H0: u 1 - u2 = 20 against H1: u 1 - u2 > 20 (exceeds 20).Then rejection region will be:
A. in Center
B. on both sides of Z=0
C. on right side of Z=0
D. on left side of Z=0

Question No:69 **(Marks:1)** **Vu-Topper RM**
The precision of an estimator can be increased by increasing the;
A. Size of sample
B. Size of population
C. Number of parameters
D. Number of estimators

Question No:70 **(Marks:1)** **Vu-Topper RM**
Pairing is done only in case of …….observations.
A. Large
B. Small
C. Dependent
D. Independent

Question No:71 **(Marks:1)** **Vu-Topper RM**
If we are testing H0 :P1 - P2 = 120 against H1 : P1 - P2 > 120 (exceeds 120).Then rejection region will be:
A. in Center
B. on left side of Z=0
C. on both sides of Z=0
D. on right side of Z=0

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Question No:72 **(Marks:1)** **Vu-Topper RM**

For a sample p1= 0.4, n1 = 100 and for second sample p2= 0.6 and n2= 100, the pooled estimator pc is:

A. 0.60
B. 0.40
C. 0.55
D. 0.50

Question No:73 **(Marks:1)** **Vu-Topper RM**

An alternative hypothesis is generally denoted by:

A. H3
B. H2
C. H1
D. H0

Question No:74 **(Marks:1)** **Vu-Topper RM**

Alpha is the probability of:

A. All
B. Type I error
C. Reject H0/H0 is true
D. Committing Type I error

Question No:75 **(Marks:1)** **Vu-Topper RM**

If H0: P = 0.50 and H1: P > 0.50 then Critical region at 5% would be:

A. Z less than and equal to1.645
B. Z greater than 1.645
C. Z greater than and equal to 1.645
D. Z equal to 1.645

Question No:76 **(Marks:1)** **Vu-Topper RM**

The median of the t-distribution is

A. Does not exist
B. Equals degrees of freedom

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C. ONE
D. ZERO

Question No:77 **(Marks:1)** **Vu-Topper RM**

If P (6.2< ? <12.2) =0.95 then

A. We are 5% confident that true value of the parameter fall in this interval
B. We are 95% confident that true value of the parameter fall in this interval
C. This is wrong statement
D. All are correct

Question No:78 **(Marks:1)** **Vu-Topper RM**

For a particular hypothesis test, Alpha=0.09 and Beta=0.03, what is the value of Type I error?

A. 0.97
B. 0.03
C. 0.09
D. 0.91

Question No:79 **(Marks:1)** **Vu-Topper RM**

The t-test for independent samples assumes which of the following?

A. Groups have equal medians
B. Scores are dependent
C. Groups have equal means
D. Scores are independent

Question No:80 **(Marks:1)** **Vu-Topper RM**

A deserving player was not selected in the cricket team. It is an example of:

A. Type-I error
B. Type-Il error
C. Type-Ill error

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D. Type-IVerror

Question No:81 **(Marks:1)** **Vu-Topper RM**

For a particular hypothesis test, Alpha=0.10 and Beta=0.15. What is the value of Type I error?

A. 0.85
B. 0.10
C. 0.90
D. 0.15

Question No:82 **(Marks:1)** **Vu-Topper RM**

Suppose H0 : u 1 = u2 and H1:u1 >u 2 and Critical value is z= 1.645. If calculated value of Z = 1.32,then what will be your conclusion?

A. Reject Ho
B. Impossible to decide
C. None of the above
D. Accept Ho

Question No:83 **(Marks:1)** **Vu-Topper RM**

The density of the t-distribution reaches its maximum at t=_____:

A. 3
B. 2
C. 0
D. 1

Question No:84 **(Marks:1)** **Vu-Topper RM**

t-distribution can be used to find the confidence interval for:

A. Population mean
B. Population variance
C. Populating proportion
D. Difference between population proportions

Question No:85 **(Marks:1)** **Vu-Topper RM**

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When sample size is to be considered large, population standard deviation can be replaced by

A. Sample Mean
B. Sample Variance
C. Population variance
D. Sample Standard Deviation

Question No:86 (Marks:1) Vu-Topper RM

If the NULL hypothesis says that the average height of Pakistani soldiers exceeds the average height of American soldiers by not more than 3 inches. What will be its ALTERNATIVE hypothesis?

A. P > 3
B. none of these
C. P < 3
D. P = 3

Question No:87 (Marks:1) Vu-Topper RM

To test the hypothesis about the difference of means for large samples, what test statistic will be used?

A. Chi-square
B. z
C. t
D. f

Question No:88 (Marks:1) Vu-Topper RM

For a particular hypothesis test, Alpha=0.10 and Beta=0.15. What is the value of Type II error?

A. 0.85
B. 0.10
C. 0.90
D. 0.15

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Question No:89 **(Marks:1)** **Vu-Topper RM**

In testing H0: $\mu = 100$ against H1: $\mu \leq 100$ at the 10% level of significance,H0 is rejected if:

A. 100 is contained in the 90% confidence interval
B. The p-value is greater than 0.10
C. The value of the test statistic is in the acceptance region
D. The p-value is less than 0.10

Question No:90 **(Marks:1)** **Vu-Topper RM**

Which of the following statements is NOT true?

A. The t distribution is more spread out than the standard normal distribution
B. As the degrees of freedom get smaller, the t-distribution's dispersion gets smaller
C. The t distribution is symmetric about zero
D. The t distribution is mound-shaped

Question No:91 **(Marks:1)** **Vu-Topper RM**

Alpha is the probability of ……..

A. Making Type I error
B. Accepting Ho when it is true
C. Rejecting H1 when it is wrong
D. Making Type II error

Question No:92 **(Marks:1)** **Vu-Topper RM**

By increasing sample size, _____ can be decreased:

A. Both Alpha and Beta
B. Beta
C. Alpha
D. None of these

Question No:93 **(Marks:1)** **Vu-Topper RM**

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The location of the critical region depends upon:

A. Value of alpha
B. Alternative hypothesis
C. Null hypothesis
D. Value of test-statistic

Question No:94 (Marks:1) Vu-Topper RM

A _____ is a range of numbers inferred from the sample that has a certain probability of including the population parameter over the long run.

A. Lower limit
B. Hypothesis
C. None of these
D. confidence interval

Question No:95 (Marks:1) Vu-Topper RM

For a sample p1 = 0.2, n1= 100 and for second sample p2= 0.5 and n2= 200, the pooled estimator pc is:

A. 0.35
B. 0.30
C. 0.40
D. 0.45

Question No:96 (Marks:1) Vu-Topper RM

All other things remaining constant, a 99% confidence interval for a population mean μ will _____ the 95% confidence interval for μ.

A. be narrower than
B. None of the above
C. be narrower than

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D. be wider than

Question No:97 **(Marks:1)** **Vu-Topper RM**

Suppose H0: p =0 H1:p < 0 and Critical value is z = - 2.33. If calculate the value of Z = - 2.60,then what will be your conclusion?

A. Impossible to decide
B. None of the above
C. Accept Ho
D. Reject Ho

Question No:98 **(Marks:1)** **Vu-Topper RM**

To test the hypothesis about the difference of means for small samples, test statistics will be used:

A. T-test
B. F-test
C. Z-test
D. All of these

Question No:99 **(Marks:1)** **Vu-Topper RM**

The t-distribution is applicable for testing about the population mean, when population standard deviation is:

A. Known
B. Unknown
C. Less than 40
D. Greater then 40

Question No:100 **(Marks:1)** **Vu-Topper RM**

If we are testing H0 : P1 < P2(or) P1 - P2 < 0 against H1 : P1 > P2(or) P1 - P2 > 0 and the tabulated value of Z = 1.645 and Z calculated value = 2.63 then what will be the conclusion: \end{gathered} \]

A. Reject Ho
B. Impossible to decide
C. Accept Ho

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D. None of the above

Question No:101 **(Marks:1)** **Vu-Topper RM**

If we have the following information, P= 0.16, q = 0.84, Z=1.96 and e = 0.05, then sample size will be:

A. 205
B. 199
C. 200
D. 207

Question No:102 **(Marks:1)** **Vu-Topper RM**

To determine a sample size in estimating population mean, we use the _____value.

A. X
B. F
C. T
D. Z

Question No:103 **(Marks:1)** **Vu-Topper RM**

If we are testing H0: P1=P2 against H1:P1 > P2(or P1 -P2 > 0 and tabulated value of z = 1.96 and Z calculated value = 1.63 then what will be the conclusion:

A. Reject Ho Reject H0
B. Accept Ho
C. Impossible to decide
D. None of the above

Question No:104 **(Marks:1)** **Vu-Topper RM**

Which one provides the basis for hypothesis testing?

A. Null hypothesis
B. Critical value
C. Test-statistic
D. alternative hypoetheis

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Question No:105 **(Marks:1)** **Vu-Topper RM**

_____ are equivalent.

A. Type-II error and Level of confidence
B. Type-I error and Level of significance
C. Type-I error and Level of confidence
D. Type-II error and Level of significance

Question No:106 **(Marks:1)** **Vu-Topper RM**

Which of the following is not a criterion for a good problem situation?

A. It must be clearly structured and defined.
B. It must be such that its solution will be benefited by group effort.
C. It must be appropriate to the students' level of development.
D. It must be authentic.

Question No:107 **(Marks:1)** **Vu-Topper RM**

The term 1- β is called

A. Level of the test
B. Power of the test
C. Size of the test
D. Critical region

Question No:108 **(Marks:1)** **Vu-Topper RM**

Critical region is also known as:

A. None of these
B. Neither a and b
C. Rejection region
D. Acceptence Region

Question No:109 **(Marks:1)** **Vu-Topper RM**

t-distribution is used to test the hypothesis about:

A. variance
B. Proportions

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C. ratio of variance
D. Mean

Question No:110 (Marks:1) Vu-Topper RM
We can apply t-distribution to test the hypothesis, when
A. Sample is large
B. Sample is small
C. Population variance is known
D. Population variance is unknown and Sample is small

Question No:111 (Marks:1) Vu-Topper RM
The critical region is computed from the value of:
A. Beta
B. Alpha
C. 1-Beta
D. 1-alpha

Question No:112 (Marks:1) Vu-Topper RM
A failing student is passed by an examiner. It is an example of:
A. Type I error
B. Type II error
C. Correct decision
D. No information regarding student exams

Question No:113 (Marks:1) Vu-Topper RM
"A point estimate plus/minus a few times the standard error of that estimate". This statement represents:
A. Critical value
B. Critical region
C. Acceptance region
D. Confidence interval

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Question No:114 **(Marks:1)** **Vu-Topper RM**

If we reject the true null hypothesis using t-distribution, we might be making:

A. Type I error
B. Type II error
C. Correct decision
D. No information regarding student exams

Question No:115 **(Marks:1)** **Vu-Topper RM**

A random sample of n=25 values gives sample mean 83. Can this sample be regarded as drawn from a normal population with μ= 80 and s= 7? In this question the alternative hypothesis will be:

A. H1: μ = 80
B. H1: μ ≠ 80
C. H1: μ > 80
D. H1: μ <80

Question No:116 **(Marks:1)** **Vu-Topper RM**

ß is the probability of:

A. Accept H0/H0 is false
B. Accept H0/H0 is true
C. Reject H0/H0 is false
D. Reject H0/H0 is true

Question No:117 **(Marks:1)** **Vu-Topper RM**

For a particular hypothesis test, α=0.05and β=0.05. The power of test is equal to:

A. 0.05
B. 0.50
C. 0.90
D. 0.95

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Question No:118 **(Marks:1)** **Vu-Topper RM**

Mean of t-distribution will not exist when d.f(degree of freedom) will equal to:

A. 4
B. 2
C. 3
D. 1

Question No:119 **(Marks:1)** **Vu-Topper RM**

To test the hypothesis about the difference of means, which test we use for testing?

A. z
B. both a and b
C. f
D. t

Question No:120 **(Marks:1)** **Vu-Topper RM**

Confidence interval is also known as

A. Level of Significance
B. None of these
C. Margin of error
D. Point estimator

Question No:121 **(Marks:1)** **Vu-Topper RM**

Accept H0/H0 is false, is the _____.

A. Type I error
B. Type II error
C. Correct decision
D. None of the above

Question No:122 **(Marks:1)** **Vu-Topper RM**

If we are testing H0 : u 1- u 2 = 10 against H1: u 1 - u 2 < 10.Then rejection region will be:

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A. on right side of Z=0
B. on both sides of Z=0
C. on left side of Z=0
D. in Center

Question No:123 (Marks:1) Vu-Topper RM
The Test-statistic is used to decide whether to_____a null-hypothesis.

A. Accept or Reject
B. Non of these
C. Accept
D. Reject

Question No:124 (Marks:1) Vu-Topper RM
Which test is preferred when you want to compare two population means where you have two samples in which observations in one sample can be paired with observations in the other sample?

A. Chi-square
B. ANOVA
C. T-test
D. Paired t-test

Question No:125 (Marks:1) Vu-Topper RM
If a significance level of 1% is used rather than 5%, the null hypothesis is:

A. Just as likely to be rejected
B. More likely to be rejected
C. Less likely to be rejected
D. None of the these

Question No:126 (Marks:1) Vu-Topper RM
For a particular hypothesis test, Alpha=0.05 and Beta=0.15. What is the level of significance?

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A. 0.85
B. 0.15
C. 0.05
D. 0.95

Question No:127 (Marks:1) Vu-Topper RM
Suppose H0: u 1= u2 and H1:u 1 < u 2 and Critical value is z= - 1.645. If calculated value of Z= - 2.08,then what will be your conclusion?
A. Accept Ho
B. None of the above
C. Reject Ho
D. Impossible to decide

Question No:128 (Marks:1) Vu-Topper RM
When examining group difference where no direction of the difference is specified, which of the following is used?
A. One-tailed test
B. Two-tailed test
C. Directional hypothesis
D. Difference of modes test

Question No:129 (Marks:1) Vu-Topper RM
The end points that bound the confidence interval are called;
A. Lower limits
B. Upper limits
C. Bounded limits
D. Lower and upper limits

Question No:130 (Marks:1) Vu-Topper RM
For a particular hypothesis test, Alpha=0.07 and Beta=0.05. What is the level of significance?
A. 0.95
B. 0.07

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C. 0.05
D. 0.93

Question No:131 (Marks:1) Vu-Topper RM
What is the rule of thumb for a sample size "n" to be considered large.
A. n<30
B. n>30 or n=30
C. n=30
D. None of the above

Question No:132 (Marks:1) Vu-Topper RM
The probability of success changes from trial to trial, is the property of:
A. Poisson experiment
B. Binomial experiment
C. Hypergeometric experiment
D. Both binomial & hypergeometric experiment

Question No:133 (Marks:1) Vu-Topper RM
The normal curve is:
A. The typical frequency curve for psychological data.
B. A convenient fiction.
C. The standard against which all other distributions are assessed.
D. A bell-shaped distribution

Question No:134 (Marks:1) Vu-Topper RM
Binomial distribution has two parameters; which are:
A. None
B. X and n
C. p and n
D. p and q

Question No:135 (Marks:1) Vu-Topper RM

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If in a lottery, one thousand tickets are sold. The chance of a ticket for wining the lottery is:

A. Mutually exclusive
B. Not equally likely
C. Equally likely
D. No information is given

Question No:136 (Marks:1) Vu-Topper RM

A sales firm receives different calls from clients. If someone says, today it will receive "at least 3 calls". How many calls will be received?

A. 0, 1, 2, 3, 5, . . .
B. 0, 1, 2, 3
C. 3, 4, 5, . . .
D. 1, 2, 3

Question No:137 (Marks:1) Vu-Topper RM

The number of trials in hypergeometric distribution is:

A. Small
B. Large
C. Not fixed
D. Fixed

Question No:138 (Marks:1) Vu-Topper RM

It is sometimes possible to obtain approximate probabilities associated with values of a random variable by using the different probability distribution of a different random variable. For example, binomial probabilities use the Poisson probability function, binomial probabilities using the normal etc. In order for the Poisson to give "good" approximate values for binomial probabilities we must have the condition(s) that:

A. The population size is large relative to the sample size
B. The probability, p, is small and the sample size is large
C. The probability, p, is close to 0.5 and the sample size is large

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D. The probability, p, is close to 0.5 and the population size is large

Question No:139 **(Marks:1)** **Vu-Topper RM**

If the second moment ratio is less than 3 the distribution will be:

A. Leptokurtic
B. Mesokurtic
C. Platykurtic
D. None of these

Question No:140 **(Marks:1)** **Vu-Topper RM**

Which of the following is a characteristic of a binomial probability experiment?

A. P(success) = P(failure)
B. Each trial has more than two possible outcomes
C. The binomial random variable x is the count of the number of trials that occur
D. The result of one trial does not affect the probability of success on any other trial

Question No:141 **(Marks:1)** **Vu-Topper RM**

Which of the following is not a property of a binomial experiment?

A. All of the above
B. the experiment consists of a sequence of n identical trials
C. each outcome can be referred to as a success or a failure
D. the probabilities of the two outcomes can change from one trial to the next

Question No:142 **(Marks:1)** **Vu-Topper RM**

If a random variable X denotes the number of heads when we toss a fair coin 5 times, the X assumed the values:

A. 0,1,2,3
B. 1, 5, 5

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C. 1, 2,3,4,5
D. 0, 1, 2,3,4,5

Question No:143 (Marks:1) Vu-Topper RM
The lottery conducted in various countries for purposes of money-making provides a good example of:
A. Binomial distribution
B. Normal distribution
C. Poisson distribution
D. Uniform distribution

Question No:144 (Marks:1) Vu-Topper RM
The number of typing errors per page in a book follows:
A. Normal Distribution
B. Poisson Process
C. Binomial Distribution
D. Hyper geometric Distribution

Question No:145 (Marks:1) Vu-Topper RM
In a binomial experiment successive trials are:
A. Both
B. None
C. Dependent
D. Independent

Question No:146 (Marks:1) Vu-Topper RM
In a binomial experiment with three trials, the variable can take:
A. 2 values
B. 4 values
C. 5 values
D. 3 values

Question No:147 (Marks:1) Vu-Topper RM

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If p is very small and n is considerably large then we shall apply the:

A. Binomial distribution
B. Hypergeometric distribution
C. Poisson distribution
D. Exponential distribution

Question No:148 (Marks:1) Vu-Topper RM

The location and shape of the normal curve is (are) determined by:

A. Mean
B. Variance
C. Mean & variance
D. Mean & standard deviation

Question No:149 (Marks:1) Vu-Topper RM

The total area under the normal curve is:

A. 0
B. 1 **Page 186**
C. 0.5
D. 0.75

Question No:150 (Marks:1) Vu-Topper RM

Poisson distribution can be used to approximate the hyper geometric distribution when; n <0.05N, n>20 and

A. p>0.05
B. p>0.10
C. p<0.05
D. p<0.10

Question No:151 (Marks:1) Vu-Topper RM

The function F(x) gives the probability of the event that X takes a value a value.....

A. P(X≤x)
B. P(X>x)

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C. P(X<x)
D. P(X=x)

Question No:152 **(Marks:1)** **Vu-Topper RM**
In hyper geometric distribution; experiment can be repeated a
A. Depends upon the situation
B. variable number of times
C. None of the above
D. fixed number of times

Question No:153 **(Marks:1)** **Vu-Topper RM**
The hypergeometric probability distribution has parameters:
A. N, n, k
B. N, N-K, n
C. N, n
D. N, P, n

Question No:154 **(Marks:1)** **Vu-Topper RM**
Mean of binomial distribution is:
A. n/p
B. np
C. n
D. p

Question No:155 **(Marks:1)** **Vu-Topper RM**
In a binomial experiment ,probability of success changes on:
A. 10th trial
B. Each trial
C. 30th trial
D. Second trial

Question No:156 **(Marks:1)** **Vu-Topper RM**

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In a bivariate probability distribution of X and Y,If f(0,1)=3/14, g(0)=5/14 and h(1)=3/7,then

A. Both of these
B. None of these
C. X and Y are statistically independent
D. X and Y are not statistically independent

Question No:157 (Marks:1) Vu-Topper RM

The values of moment ratios b1 and b2 of normal distribution are:

A. 0 and 1
B. 0 and 2
C. 0 and 3
D. 0 and 4

Question No:158 (Marks:1) Vu-Topper RM

If value of p is smaller or lesser than 0.5 then binomial distribution is classified as:

A. skewed to left
B. skewed to infinity
C. skewed to right
D. skewed to integers

Question No:159 (Marks:1) Vu-Topper RM

E(X + Y) = ?

A. E(X + Y) = E(X) E(Y).
B. E(X + Y) = E(X) + E(Y).
C. E(X + Y) = E(X) + E(Y).
D. None

Question No:160 (Marks:1) Vu-Topper RM

The mean of Uniform distribution defined on the interval [a,b] is:

A. (a−b)
B. a+b2

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C. (a+b)
D. a−b2

Question No:161 **(Marks:1)** **Vu-Topper RM**
In a normal distribution how much area lies between μ±σ
A. 80%
B. 65%
C. 68%
D. 75%

Question No:162 **(Marks:1)** **Vu-Topper RM**
We can apply binomial distribution formula, when we are drawing a sample from finite population without replacement and the sample size "n" is not more than ……..% of the population size N.
A. 8%
B. 5%
C. 10%
D. 1%

Question No:163 **(Marks:1)** **Vu-Topper RM**
For good approximation of Poisson distribution to the binomial distribution, which of the following condition (s) is/are required:
A. The population size is large relative to the sample size
B. The probability, p, is small and the sample size is large
C. The probability, p, is close to .5 and the sample size is large
D. The probability, p, is close to .5 and the population size is large

Question No:164 **(Marks:1)** **Vu-Topper RM**
Poisson distribution is never:
A. Symmetrical
B. Asymmetrical
C. Negatively skewed
D. Positively Skewed

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Question No:165 **(Marks:1)** **Vu-Topper RM**

If X and Y are independent, then Cov (X, Y) is equal to:

A. 0

B. 1

C. None

D. 0 and 1

Question No:166 **(Marks:1)** **Vu-Topper RM**

In a binomial experiment:

A. None of the above

B. Number of trials "n" is a fixed

C. Number of trials "n" is a variable

D. Number of trials "n" is not specified

Question No:167 **(Marks:1)** **Vu-Topper RM**

The location parameter of normal distribution is:

A. Variance

B. Mean

C. Range

D. Standard Deviation

Question No:168 **(Marks:1)** **Vu-Topper RM**

If total probability of joint probability distribution f(x,y) is ONE then what will be the total probability of its marginal pdf h(y)?

A. One

B. Less than one

C. Depend on given values

D. Anywhere between zero and one

Question No:169 **(Marks:1)** **Vu-Topper RM**

The hypergeometric random variable is a(an):

A. Undefined

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B. Independent
C. Discrete variable
D. Continuous variable

Question No:170 (Marks:1) Vu-Topper RM
The correlation coefficient:
A. makes the scatterplot redundant.
B. should be interpreted alongside the relevant scatterplot.
C. all of these.
D. can be expressed as a percentage.

Question No:171 (Marks:1) Vu-Topper RM
Suppose 60% of a herd of cattle is infected with a particular disease. Let Y = the number of non-diseased cattle in a sample of size 5. the distribution of Y is:
A. Poisson with u = .6
B. Binomial with n = 5 and p = 0.4
C. Binomial with n = 5 and p = 0.5
D. Binomial with n = 5 and p = 0.6

Question No:172 (Marks:1) Vu-Topper RM
Density function of normal distribution attains its maximum value when the value of random variable X is:
A. X = Standard Deviation
B. X = Mean
C. All above
D. X = Variance

Question No:173 (Marks:1) Vu-Topper RM
The mean of the continuous probability function f(x) can be obtained by
A. Multiplying the x by its function f(x)
B. Integrating the product xf(x)
C. Integrating the function f(x)

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D. Integrating the x

Question No:174 **(Marks:1)** **Vu-Topper RM**

In a bivarite probability distribution of X and Y,If E(X)=3 and E(Y)=5, then E(X-Y)= ?

A. -2
B. 15
C. None of these
D. 2

Question No:175 **(Marks:1)** **Vu-Topper RM**

In case of bivariate distribution, co-variance of distribution is represented:

A. E(X.X)-E(X).E(Y)
B. E(Y.X)-E(X).E(Y)
C. E(Y.Y)-E(X).E(Y)
D. E(X^2)-[E(X)]^2

Question No:176 **(Marks:1)** **Vu-Topper RM**

What will be the value of C.V if S.D = 0.5 and mean = 1.

A. 50%
B. 45%
C. 30%
D. 10%

Question No:177 **(Marks:1)** **Vu-Topper RM**

Which of the following is the property of hypergeometric experiment?

A. Sampling is performed without replacement
B. p remains constant from trial to trial
C. n is not fixed
D. Successive trials are independent

Question No:178 **(Marks:1)** **Vu-Topper RM**

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If total probability of joint probability distribution f(x,y) is ONE then what will be the total probability of its marginal pdf g(x)?

A. One
B. Half of the joint pdf
C. Anywhere between zero and one
D. Less than one

Question No:179 **(Marks:1)** **Vu-Topper RM**

If the first moment ratio is equal to 0, then the distribution will be:

A. Positively Skewed
B. Symmetrical
C. Leptokurtic
D. Negatively Skewed

Question No:180 **(Marks:1)** **Vu-Topper RM**

For the Poisson distribution P(X = 1) = e−2211! the mean value is:

A. 0
B. 2
C. 1
D. 3

Question No:181 **(Marks:1)** **Vu-Topper RM**

The shape parameter of normal distribution is:

A. Standard Deviation
B. Median
C. Mean
D. Mode

Question No:182 **(Marks:1)** **Vu-Topper RM**

In hyper geometric distribution probability of success always:

A. None of the above
B. sometimes varies and sometimes is constant
C. Varies from trial to trial

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D. is constant from trial to trial

Question No:183 **(Marks:1)** **Vu-Topper RM**

The number of parameters in Poisson distribution is (are):

A. 0
B. 1
C. 2
D. 3

Question No:184 **(Marks:1)** **Vu-Topper RM**

Using the normal approximation to the binomial distribution with n= 3 and p= 0.0571 the value of mean is:

A. 0.5133
B. 0.1713 **Page 235**
C. 0.2132
D. 0.1923

Question No:185 **(Marks:1)** **Vu-Topper RM**

Which type of data is used in the binomial distribution?

A. Continues data only
B. Both discrete and continuous
C. Discrete data only
D. Qualitative data only

Question No:186 **(Marks:1)** **Vu-Topper RM**

The relationship between the mean and variance of Poisson distribution is:

A. Mean >Variance
B. Mean = Variance
C. Mean <Variance
D. Mean = Standard deviation

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Question No:187 **(Marks:1)** **Vu-Topper RM**

In a normal distribution how much area lies between $\mu \pm 2\sigma$

A. 65%
B. 75%
C. 95%.45%
D. 94%

Question No:188 **(Marks:1)** **Vu-Topper RM**

A correlation coefficient:

A. all of these.
B. is a sort of index of how close the points of a scattergram deviate from the best fitting straight line through those points.
C. tells you the direction of the slope of the scattergram.
D. efficiently summarises some of the information in a scatterplot.

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